

MultiPlanetary Exchange

Final Design Paper - Ceres Clearing House

Design thesis. Keep local markets fast; batch cross-planet orders fairly; make custody unique; pre-fund remote risk; make finality and recovery explicit.

Executive summary

The exchange serves all nine settlements through local continuous markets and a Ceres-hosted global batch market. Each unit of cash or shares has one authoritative settlement. Remote orders reserve assets before network dependence, and retries are idempotent. Ceres also clears the implemented cash-settled futures product.

Objective	Mechanism
Asset safety	Unique authority; reservation; durable transaction IDs; explicit finality.
Fairness	Cross-planet batch: price -> source timestamp -> pro-rata exact ties.
Local liveness	Same-settlement trades and transfers continue during remote outages.
Solvency	Delay-aware margin, maintenance calls, funded default waterfall.
Auditability	Durable states, source timestamps, immutable batch inputs, reconciliation.

1. Scope and implementation status

Implemented and demonstrated: equity/share trading and transfers; local and global matching; clearing, settlement, margin, default, retries, disconnection, and ordinary recovery/deduplication; and a cash-settled MOI futures framework. Shifted difficult-epoch evidence exposes one known implementation gap: sessions do not yet proactively re-pin and resubmit abandoned financial messages when a pinned route later closes. Proactive re-pinning/resubmission is the production design rule, not a capability claimed by the current demo.

Product / function	Status	Rule
Equities / shares	IMPLEMENTED	Continuous locally; cross-planet via Ceres batch; delivery-versus-payment settlement.
Cash transfers	IMPLEMENTED	Authoritative local ledgers; remote transfer uses reservation and destination validation.
Cash-settled futures	IMPLEMENTED	MOI future; Ceres margin, mark-to-market, maturity, default.
Options / bonds / currencies / lending / shorts	EXTENSION	May reuse custody/clearing framework, but not claimed as

Product / function	Status	Rule
		demonstrated support.

2. Architecture and responsibilities

Actor	Knows / controls	Responsibility
Client / home operator	Local balances, own instructions, received market data	Timestamp and authenticate instructions; reserve local assets; display pending/final states.
Settlement ledger	Authoritative local cash/shares	Prevent double use; maintain locks; commit/reconcile transfers.
Ceres batch market	Eligible cross-planet orders	Freeze batch input; match by published rule; publish allocations.
Ceres clearing	Futures positions, collateral, MOI observations	Margin, maturity, default, guarantee fund.
Relay/network service	Packets/sessions only	Transport; ACK/retry/expiry. A network ACK is not financial acceptance.

Financial state machine. CREATED -> RESERVED -> SENT -> RECEIVED -> VALIDATED -> COMMITTED -> FINAL. Side states: REJECTED, EXPIRED, DEFAULT, RECOVERY.

3. Local versus remote operation

Action	When connected	If required remote authority is unavailable
Same-settlement share trade	Execute continuously	Continue.
Local cash transfer	Execute locally	Continue.
Local cancellation	Execute locally	Continue.
Cross-planet trade / transfer	Batch + backbone	Remain pending/queued or reject before commitment.
New futures position	Ceres validates funded margin	Pause; do not open.

Uncertainty never creates free assets: if a financial outcome may already have been accepted remotely, the reserved resource stays locked until reconciliation resolves the same transaction ID.

4. Global batch auction

Rule	Final design
Eligibility	Different-settlement share trades enter the global batch; same-settlement trades stay continuous.
Batch length	110% of the longest expected one-way delivery time among eligible settlements at batch start.

Rule	Final design
Transport allowance	Three hop retries / four total launches where the backbone service permits them.
Priority	Best price; equal price -> authenticated source timestamp; exact timestamp tie -> pro-rata.
Clearing price	Midpoint of the tied executable range; never violates a participant limit.
Late order	Roll to next batch if still valid; preserve its limit.
Cancellation	Effective only if received before batch execution.

Fairness example. Neptune BUY 10 @ \$100 stamped 10:00 beats Earth BUY 10 @ \$100 stamped 10:01 even if Earth's packet reaches Ceres first.

Auction execution sequence

Step	Ceres batch-market action	Economic effect
1. Admit	Accept only authenticated, funded/reserved cross-planet orders received before the batch cutoff; retain each home operator's receipt stamp.	No unfunded order can compete; distance cannot rewrite the source time.
2. Freeze	At cutoff, make the eligible order set immutable for that batch. Late valid orders roll to the next batch with the same limit.	Everyone can recompute the same input set; a late packet cannot displace an already eligible order.
3. Rank	Sort buys by higher price first and sells by lower price first. At the same price use authenticated source receipt time; exact timestamp ties are pro-rata.	Price protection comes first; propagation delay is not a tie-breaker.
4. Match	Choose the maximum executable quantity where buy limits meet sell limits. Partial fills are allowed; unmatched reservations are released after the batch result is final.	Maximizes executable volume without crossing a participant's stated limit.
5. Price	When a tied executable range exists, clear at its midpoint. Example: \$45 sell vs \$50 buy clears at \$47.50, splitting the \$500 surplus \$250/\$250 for 100 shares.	Deterministic and side-neutral in the tested comparison.
6. Settle	Publish signed allocation/result	Ceres determines the trade;

	records. The matched cash/share legs then settle directly between authoritative home operators rather than detouring through Ceres.	custody ledgers determine final ownership.
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The equal-price test is deliberately adversarial: Earth, Mars, and Neptune remote bidders each submit at \$100 one minute before C-Finn at Ceres. Their packets reach Ceres roughly 0.5 h, 0.5 h, and 4.1 h later, respectively. Source-stamp priority still gives the earlier remote bidder the shares; arrival-time priority incorrectly gives C-Finn all three ties. The engine's equal-price auction passes 6/6 checks, while the deliberately substituted arrival-time rule fails three checks.

The batch transport allowance is also a fairness mechanism, not merely a latency choice. At the tested geometry, Neptune's slowest hop has a 42.7% launch-loss probability. Waiting through all three permitted retries produces a roughly 30.7 h batch and reduces a same-batch miss to the four-launch hop-abandonment case (~3.3%); using only one retry would leave an 18.2% same-batch miss probability for that hop.

5. Custody, settlement, and finality

Stage	Source	Destination
Reserve	Transferred amount locked; remainder usable	No authority yet
Send	Commitment travels under transaction ID	Awaiting validation
Validate	Source remains irrevocably bound	Destination becomes authoritative; asset is spendable
Reconcile	Source learns final state; closes lock record	Already final

- Transport receipt does not equal financial acceptance.
- Duplicate financial messages return the existing result and cannot execute twice.
- Packet expiry stops forwarding but does not cancel an unresolved financial obligation.
- After destination finality, a lost return confirmation cannot undo the transfer.

5.1 Routing policy and path selection

Routing separates control traffic from settlement traffic. Orders, batch results, margin instructions, and other authority-bound messages use the fixed Relay A/B backbone toward Ceres or the relevant authority. Once an equity match is known, the two custody operators settle the cash/share legs directly over their pre-established operator session when available. This avoids forcing every asset movement through the clearing hub.

Routing decision	Rule	Why it is used
Authority destination	Send the instruction to the node that can legally/economically decide it: Ceres batch for global matching, Ceres Clearing for futures, Mars for MOI	A relay is transport only; it never becomes financial authority.

	observations, or the authoritative custody ledger for settlement.		
Relay choice	Use the currently usable Relay A/B path. Production rule: re-pin before predictable blockage and resubmit unresolved financial messages under the same transaction ID. Current demo limitation: this proactive re-pin/resubmit path is not yet implemented for every shifted-epoch closure.	E4 shows an alternate route exists across refined closure boundaries; E5 shows why automatic route migration is necessary for liveness.	
Retry behavior	A failed backbone hop may use the service's four total launches. Retries preserve the financial transaction ID; a transport retry is not a new order or transfer.	Prevents duplicate economic actions and makes loss recovery auditable.	
Settlement legs	After matching, prefer direct operator-to-operator sessions for the actual cash/share legs.	Measured examples are faster and use one shared-quota packet instead of two through Ceres.	
No usable route	Keep already-committed resources funded/locked and mark the operation pending; reject or queue work that has not committed. Packet expiry alone never cancels an obligation.	Safety is preserved through partitions; no asset becomes spendable twice.	
Route restoration	Reconcile by durable transaction ID, deduplicate, learn destination finality, then release only the correct residual lock.	Lost confirmations cannot reverse an already final destination credit.	
Settlement leg	Direct session	Via Ceres	Quota packets
Earth -> Mars	0.6 h	1.1 h	1 direct vs 2 relayed
Earth -> Neptune	4.2 h	4.7 h	1 direct vs 2 relayed
Mars -> Neptune	4.1 h	4.6 h	1 direct vs 2 relayed

The direct-settlement choice costs six sessions established before hour 0 (one SYN packet each, permitted from hour -168 in the team evidence). It then saves a relay hop and half the shared-quota packet count on the measured settlement legs. This routing rule is independent of the decision to keep Ceres as the market/clearing hub: Ceres decides global matches and derivatives obligations, while authoritative endpoint ledgers move the resulting assets.

6. Futures, valuation, margin, and default

Term	Rule
Product	Cash-settled Mars Ore Index (MOI) future; no physical delivery.
Price source	Mars; signed observation every 12 model hours with source timestamp.
Maturity	300 model hours; first scheduled MOI observation at/after maturity is final settlement price.
Initial margin	5% per observation in the implemented demo, sized/checked against modeled adverse movement and communication risk.
Maintenance	75% of required initial-margin allowance.
Margin deadline	Per-location communication window: round trip to Ceres plus one retry opportunity.
Default	If funded top-up is not validated by Ceres by deadline, close at next authoritative MOI observation.
Waterfall	Defaulter margin -> extra collateral -> pre-funded \$50,000 Ceres guarantee fund.

The same margin rule is fixed before the rising and falling price paths are run; it may react only to information already published and received.

7. Opening balance sheet and funding

Account	Settlement	Cash	Shares
E-Alice	Earth	\$70,000	600
E-Bob	Earth	\$45,000	400
M-Carla	Mars	\$60,000	550
M-Diego	Mars	\$45,000	450
N-Eve	Neptune	\$55,000	500
C-Finn	Ceres	\$35,000	500
TOTAL		\$310,000	3,000

Guarantee fund funding after hour 0. Ceres Clearing starts with no institutional endowment. E-Alice and E-Bob later contribute \$25,000 each, creating the \$50,000 guarantee fund used by the futures default waterfall. The opening balance sheet therefore totals \$310,000, not \$360,000. The balance sheet uses 6 named accounts across 4 settlements and remains below the brief's \$500,000 / 5,000-share limits. Reserved cash, shares, collateral, and guarantee resources are explicitly encumbered and cannot support another obligation.

8. Service table - all nine settlements

Service is conditional on the required authority being reachable. Local-safe actions remain available without Ceres. Cross-planet service is asynchronous and may wait through closures; no unconditional completion deadline is promised.

Settlement	Equity trading	Cash transfer	MOI futures	Main restriction
Mercury	Local continuous; remote batch	Local / remote	Open via Ceres	Remote completion depends on route + batch.
Venus	Local continuous; remote batch	Local / remote	Open via Ceres	Remote completion depends on route + batch.
Earth	Local continuous; remote batch	Local / remote	Open via Ceres	Ceres hub adds backbone traffic for Earth-heavy accounts.
Mars	Local continuous; remote batch	Local / remote	Open via Ceres	MOI originates here; futures still clear at Ceres.
Ceres	Local continuous; global batch local	Local / remote	Clearing local	Hub-local access does not get arrival-time priority.
Jupiter	Local continuous; remote batch	Local / remote	Open via Ceres	Longer routes raise delay/loss.
Saturn	Local continuous; remote batch	Local / remote	Open via Ceres	Longer routes raise delay/loss.
Uranus	Local continuous; remote batch	Local / remote	Open via Ceres	High remote delay/loss; batch remains fair.
Neptune	Local continuous; remote batch	Local / remote	Open via Ceres	Poorest remote service; often determines batch length.

9. Ceres clearing-house decision

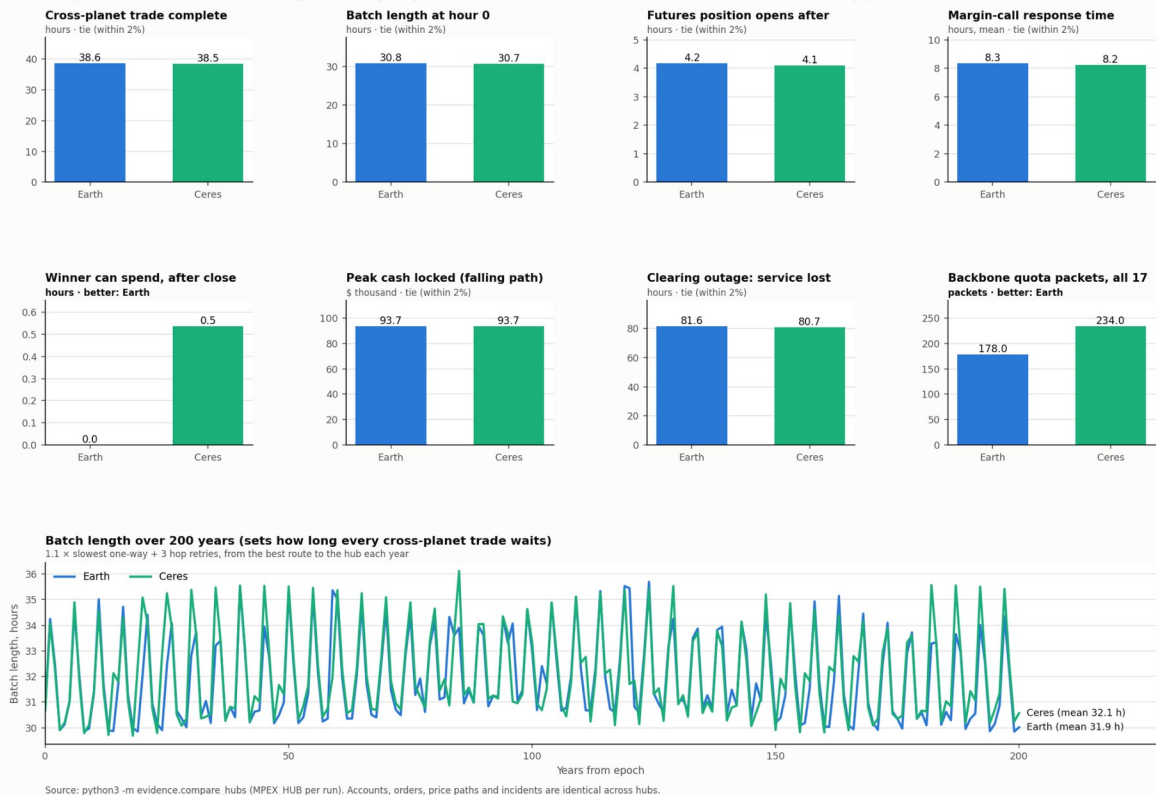
Metric	Earth hub	Ceres hub	Effect
Cross-planet completion	38.6 h	38.5 h	Tie
Mean batch length, 200 y	31.9 h	32.1 h	Tie
Time with one relay	8.6%	0.8%	Ceres better

Metric	Earth hub	Ceres hub	Effect
blocked			
Planned re-routes, 200 y	316	4	Ceres better
Longest observed one-relay spell	21 d	190 d	Earth better
Backbone quota, 17 scenarios	178	234	Earth better

Ceres is selected because market speed is effectively unchanged while the hub spends much less time exposed to a one-relay state. Accepted costs are about 31% more backbone quota with today's Earth-heavy balance sheet and rare long Ceres blockage spells. Proactive re-pinning before predicted long blockages is part of the production design, but it is not yet implemented for every shifted-epoch closure; E5 demonstrates this liveness limitation.

Clearing house at Earth vs Ceres: the same 17 scenarios, rerun

Every number below comes from running the exchange engine on the network model with the hub moved; lower is better in every panel.



Earth vs. Ceres rerun on the same 17 exchange scenarios. Lower is better in the displayed panels.

10. Failure handling and exact recovery

Failure	What each actor does / owes
Packet lost	Transport may retry. Financial layer keeps the same transaction ID; no new economic instruction is created.

Failure	What each actor does / owes
Confirmation lost	Sender does not unlock; it queries/reconciles the existing transaction.
Duplicate	Receiver returns existing financial result; balances do not change twice.
Contradictory records	Authoritative custody ledger + durable transaction state decide; disputed batch recomputed from immutable inputs.
No route / expired packet	Remote work stays unresolved and funded. Expiry never cancels the obligation.
Ceres isolation	Local-safe service continues; new futures pause; unresolved clearing obligations remain funded.

Recovery order in the design: restore/re-pin sessions -> exchange durable state summaries -> deduplicate -> reconcile authority/locks -> resubmit unresolved financial messages under the same transaction ID -> process overdue maturity and margin events -> reopen global services. The current implementation performs ordinary recovery/deduplication but E5 shows that proactive re-pinning/resubmission is incomplete at a difficult Ceres-relay epoch; a funded future can therefore remain open past maturity.

11. Risks, limits, and treatments

Rank	Risk	Treatment / accepted cost
1	Pinned-session liveness gap at difficult shifted epochs	E5: a future remains funded/open past maturity when its pinned Mars->Ceres path closes and abandoned marks are not resubmitted. Production fix: proactive re-pin + transaction-ID resubmission.
2	Neptune dominates batch timing and has high launch loss	Batch deadline includes transport allowance; incomplete orders roll/pause rather than receive unfair priority.
3	Ceres hub increases traffic for Earth-heavy balances	Accepted +31% in current 17-scenario comparison; move collateral/accounts only if economics justify it.
4	Lost confirmation can lock capital for a long time	Safety over liquidity: keep funded lock, reconcile by transaction ID.
5	Central clearing outage pauses global derivatives	Local markets continue; no new under-margined futures; recovery is explicit.

Guarantees are conditional on the stated protocol and the fixed network model. Numerical scans are evidence about the modeled horizon, not proofs of perpetual connectivity.

12. Long-term operation - no reset

Requirement	Rule that continues indefinitely
Long no-route period	Safety: obligations remain funded/pending and local-safe service continues. Liveness after restoration requires re-pin/resubmission; E5 identifies this as incomplete in the current demo.
Backlog after reopen	Process durable queues by product rule and source time; enforce current quota/capacity rather than bursting without limit.
Very late maturity	Contract stores maturity rule and funded owner until authoritative price arrives; maturity is not tied to scenario end.
30-day packet expires	Protocol requirement: commitment persists and later resubmission uses the same financial transaction ID. Current demo does not yet automate this in every difficult shifted-epoch case.
Capital replenishment	Only earned transfers or explicit contributions replenish capital; no free future endowment.
Identifiers / storage	128-bit financial IDs; never reuse an ID. Archive finalized records after a bounded audit period; unresolved records remain live.
Quota enforcement	Both direct and backbone quotas are checked before originating new application traffic; automatic retries follow the brief.
Demand limit	Admission control: when funded demand exceeds available quota/capital, new remote work waits or is rejected; no unlimited-throughput promise.

Infrastructure assumption: the nine settlement gateways, Relay A/B network, local ledgers, authenticated clocks/keys, and durable storage exist. The design adds no relay or bandwidth beyond the brief. Financial transaction IDs remain durable across packet expiry; however, the current demo does not yet automatically resubmit an abandoned maturity mark on a newly usable route.

13. Deployment effects outside the baseline

Effect	Numeric estimate + source/assumption	Effect on guarantees
Ceres local rotation / ground visibility	NASA: Ceres rotates in about 9 h. A single equatorial surface	Baseline assumes settlement gateway access. Real

Effect	Numeric estimate + source/assumption	Effect on guarantees
	antenna with a zero-degree horizon and a slowly moving relay direction could be geometrically hidden for roughly half a rotation (~4.5 h).	deployment needs multiple sites, an orbiter, or store-and-forward. Financial safety unchanged; availability promise becomes conditional on gateway visibility.
Planet-local timekeeping	NASA: Mars solar day = 24 h 39 m 35 s; ~39.6 min longer than an Earth day. A naive same-local-clock daily schedule drifts by ~24 h in ~36.4 Earth days.	Use model elapsed seconds/TDB for financial timestamps and convert only for display. No change to priority/finality if clocks are authenticated and converted correctly.

Sources: NASA Science, 'Ceres: Facts' (rotation ~9 h); NASA GISS Mars24 technical notes / JPL Mars facts (Mars sol 24 h 39 m 35 s). These effects are not part of the common baseline model and are not used to improve the scored baseline numbers.

14. Evidence and project-compliance map

The design paper states the rules and explicitly separates intended production behavior from demonstrated implementation behavior. The accompanying evidence appendix carries S1-S3 and E1-E5 calculations, including the difficult-epoch limitation rather than hiding it.

Project requirement	Where addressed
Architecture + operating rulebook	Sections 2-6
Supported products + implemented/extension status	Sections 1 and 6
Responsibilities and actor knowledge	Sections 2, 5, 10
Guarantees, conditions, restrictions, service table	Sections 3, 8, 15
Risks and limits	Section 11
Long-term operation	Section 12
Deployment assessment with two numeric estimates	Section 13
Opening balance sheet	Section 7
Scenarios / tests	Section 14 summary; detailed traces belong in evidence appendix
S1-S3, E1-E5	Separate 12-page evidence appendix required by the brief

15. Guarantees and costs

We guarantee under the protocol	Condition / cost
No duplicate spendable ownership	Authoritative custody and durable locks are

We guarantee under the protocol	Condition / cost
	available.
No order executes outside its limit	Batch input is valid and immutable.
Equal-price global priority is not based on network arrival	Authenticated source timestamps; exact ties are pro-rata.
Local-safe service during remote disconnection	Required assets/authority are local.
Remote uncertainty does not create free assets	Capital may remain locked until reconciliation.
Futures losses are funded to the declared waterfall	Admission control rejects positions that cannot be funded.

Conclusion: the design makes interplanetary delay economically manageable through local activity, fair global batching, pre-funded risk, unique custody, and explicit finality. The evidence supports the safety/accounting rules and broad routing availability, while also identifying a concrete liveness limitation in the current demo: difficult shifted-epoch route closure can leave a funded future open until proactive re-pinning/resubmission is implemented.

References

1. MultiPlanetary Exchange System - Participant Brief, released 2 Oct 2026 (common project rules and evidence requirements).
2. NASA Science, Ceres: Facts - Ceres rotation about 9 hours.
3. NASA GISS Mars24 / JPL Mars facts - mean Mars solar day 24 h 39 m 35 s.
4. Team evidence: hub-earth-vs-ceres comparison, design-choice experiments, and 17-scenario exchange test suite.